# A tetrahedrally symmetric four-component link whose three-component sublinks are Borromean

Neeraja Abhyankar

October 12, 2026

**Abstract**

We exhibit a link of four components in $\mathbb{R}^3$ with two properties: every one of its four three-component sublinks is the Borromean rings, and the embedding is carried onto itself by all twelve rotations of a regular tetrahedron. The components are given by twenty-four points with integer coordinates, generated from two of them by the symmetry group. We also prove two obstructions that explain why such a link must look complicated: no configuration of four *convex planar* components can have the required linking, and neither can any configuration in which each pair of components crosses only the minimum possible number of times.

## 1 Introduction

The Borromean rings are three closed loops, no two of which are linked, which nevertheless cannot be pulled apart; cut any one and the other two fall free. This paper is about a four-component analogue in which the Borromean property holds for every triple at once.

**Problem 1.1.** Find a link $L = K_1 \sqcup K_2 \sqcup K_3 \sqcup K_4$ in $\mathbb{R}^3$ such that

(i) every three-component sublink of $L$ is the Borromean rings, and

(ii) $L$ is carried onto itself by a group of rotations isomorphic to the rotation group of a regular tetrahedron, acting faithfully on the components.

Our main result is that such a link exists, together with an explicit model (Theorem 6.1). Two features of the problem make it more rigid than it first appears, and both are proved below: condition (i) already forces every pair of components to be unlinked (Lemma 3.1), and condition (ii) already forces the symmetry to permute the four components transitively (Lemma 3.2). So the problem really has two hypotheses rather than the four one might be tempted to impose.

The combinatorial idea the construction rests on is due to Kyle Miller [6]: place one component on each *face* of a tetrahedron. A tetrahedron has four faces and four vertices, and the three faces meeting at a vertex are exactly the three that avoid the opposite face. Choosing three of the four components is therefore the same as naming a vertex, and condition (i) becomes the statement that the components weave in the Borromean pattern around each of the four vertices. Since the symmetry group is transitive on vertices, the four instances of condition (i) are a single condition. This is what makes the problem tractable.

Sections 2–4 set up the problem and the equivariant frame. Section 5 checks that the symmetry is compatible with the linking that condition (i) demands, which is a necessary condition for the problem to have any solution at all. Section 6 gives the link.

## 2 Conventions

A *link* of $n$ components is a smooth embedding of $n$ disjoint circles in $\mathbb{R}^3$; we write $L = K_1 \sqcup \cdots \sqcup K_n$ and call the $K_i$ its *components*. Two links are *equivalent* if some ambient isotopy of $\mathbb{R}^3$ carries one onto the other respecting the labelling of components. For $S \subseteq \{1, \dots, n\}$, the *sublink* $L_S$ is $\bigcup_{i \in S} K_i$. The *unlink* $U_n$ is $n$ components bounding disjoint embedded discs.

We write $\mathcal{B}$ for the *Borromean rings*: the three-component link L6a4 of the standard tables [5], whose standard diagram is Figure 1. It has the two properties quoted above, namely $\mathrm{lk}(K_i, K_j) = 0$ for each pair while $\mathcal{B}$ is not the unlink. The diagram is drawn with circular arcs, but no three *round circles* in space form the Borromean rings: that is a theorem of Freedman and Skora [1], with an elementary proof due to Lindström and Zetterström [4]. Any realisation therefore deforms the circles, a constraint the present problem inherits.

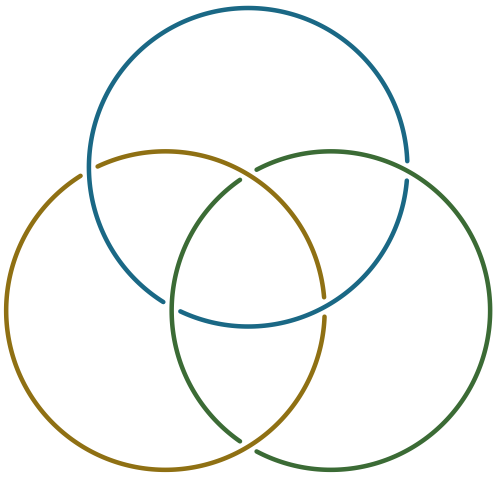


Figure 1: The standard diagram of the Borromean rings $\mathcal{B}$. Each component passes over one neighbour and under the other, cyclically. No two of the three are linked, yet all three together cannot be separated. Here and throughout, over and under are shown only by the break in the lower strand; the colours distinguish components and carry no other information.

The *linking number* $\mathrm{lk}(K_i, K_j) \in \mathbb{Z}$ of two oriented components is half the sum of the signs of the crossings between them in any diagram; it is an invariant of the link, and it vanishes for every pair of $\mathcal{B}$. Where a finer measure is needed we use Milnor's triple invariant. If all the pairwise linking numbers of a link vanish, then for distinct $i, j, k$ the invariant $\bar{\mu}(ijk) \in \mathbb{Z}$ of [7, 8] is defined with no indeterminacy, it is unchanged by an orientation-preserving diffeomorphism of $\mathbb{R}^3$, and it is alternating: interchanging two of its three indices reverses its sign. For the Borromean rings $\bar{\mu}(123) = \pm 1$, the sign depending on the chosen orientations and diagram.

Finally, let $\mathrm{T} \leq SO(3)$ denote the rotation group of a regular tetrahedron: the identity, eight rotations through $\pm 120°$ about the four axes joining a vertex to the centre of the opposite face, and three rotations through $180°$ about the axes joining midpoints of opposite edges, so $|\mathrm{T}| = 12$ and $\mathrm{T} \cong A_4$. An embedded link $L \subset \mathbb{R}^3$ is $\mathrm{T}$-*symmetric* if $g(L) = L$ as a set for every $g \in \mathrm{T}$; this induces a homomorphism $\rho \colon \mathrm{T} \to S_n$ by $g(K_i) = K_{\rho(g)(i)}$. Note that symmetry is a property of an embedding and not of an equivalence class: a link type "has tetrahedral symmetry" when some representative embedding does.

## 3 The problem, and two things one need not assume

**Lemma 3.1.** *If every three-component sublink of $L$ is $\mathcal{B}$, then $\mathrm{lk}(K_i, K_j) = 0$ for every pair, and indeed every two-component sublink of $L$ is the unlink $U_2$.*

*Proof.* Any two-element subset of $\{1,2,3,4\}$ is contained in a three-element subset $S$. By hypothesis $L_S \cong \mathcal{B}$, and every two-component sublink of $\mathcal{B}$ is $U_2$. Sublinks of equivalent links are equivalent, so the two-component sublinks of $L$ are each $U_2$. □

**Lemma 3.2.** *Let $L$ have four components and let $\rho\colon \mathrm{T} \to S_4$ be injective. Then $\rho(\mathrm{T}) = A_4$, the action on components is transitive, and the stabiliser of each component is cyclic of order 3, generated by a rotation through $120^\circ$ about one of the four vertex-to-face axes.*

*Proof.* $\rho(\mathrm{T})$ is a subgroup of $S_4$ isomorphic to $A_4$, hence of order 12 and of index 2; the only such subgroup of $S_4$ is $A_4$ itself. The action of $A_4$ on four points is transitive, so by orbit–stabiliser each stabiliser has order 3. The elements of order 3 in T are precisely the rotations through $\pm 120^\circ$ about the four vertex-to-face axes. □

So Problem 1.1 needs only the two conditions stated. In particular "pairwise unlinked" is not an independent requirement, and transitivity on components is a consequence rather than a hypothesis.

## 4 The tetrahedral frame

Fix the regular tetrahedron $\Delta$ with vertices

$$v_1 = (1,1,1), \quad v_2 = (1,-1,-1), \quad v_3 = (-1,1,-1), \quad v_4 = (-1,-1,1),$$

and let $F_i$ denote the face of $\Delta$ opposite $v_i$. By Lemma 3.2 we may label the components so that the stabiliser of $K_i$ is generated by the rotation $r_i$ through $120^\circ$ about the axis through $v_i$; we then say $K_i$ *belongs to* $F_i$. In these coordinates $r_1$ is the cyclic permutation of coordinates

$$r_1(x,y,z) = (z,x,y), \tag{1}$$

and the three half-turns about the coordinate axes,

$$u_2(x,y,z) = (x,-y,-z), \qquad u_3(x,y,z) = (-x,y,-z), \qquad u_4(x,y,z) = (-x,-y,z), \tag{2}$$

satisfy $u_i(v_1) = v_i$ and so represent the three nontrivial cosets of $\langle r_1 \rangle$ in T.

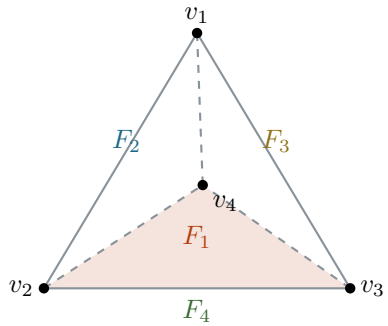
**Lemma 4.1** (Generation principle)**.** *Let $K \subset \mathbb{R}^3$ be any closed curve with $r_1(K) = K$, and set $K_1 = K$, $K_i = u_i(K)$ for $i = 2,3,4$. If these four curves are pairwise disjoint, then $L = K_1 \sqcup K_2 \sqcup K_3 \sqcup K_4$ is* T*-symmetric with $\rho$ injective.*

*Proof.* Write $u_1 = \mathrm{id}$. For $g \in \mathrm{T}$ and $i \in \{1,2,3,4\}$ there are a unique $j$ and a unique $m \in \{0,1,2\}$ with $g\,u_i = u_j r_1^m$, because the $u_i$ represent the cosets of $\langle r_1 \rangle$. Hence $g(K_i) = g\,u_i(K) = u_j r_1^m(K) = u_j(K) = K_j$, so $g$ permutes the components. The resulting permutation determines $g$ on cosets, and since T acts faithfully on $\mathrm{T}/\langle r_1 \rangle$, $\rho$ is injective. □

Lemma 4.1 is what makes the symmetry exact rather than approximate: one designs a single $r_1$-invariant curve and the group produces the rest, so there is no step at which the symmetry could be lost.

**Lemma 4.2** (Vertices and triples)**.** *The map $S \mapsto v_k$, where $\{k\} = \{1,2,3,4\} \setminus S$, is a bijection from three-element subsets of $\{1,2,3,4\}$ to vertices of $\Delta$, and it is equivariant for the actions of* T*. The components belonging to the three faces meeting at $v_k$ are exactly $\{K_i : i \neq k\}$.*

*Proof.* The three faces through $v_k$ are those containing $v_k$, namely $F_i$ for $i \neq k$, since $F_k$ is the face opposite $v_k$. Complementation is a bijection and commutes with the induced action. □




| drop component | weave at vertex |
|---|---|
| $K_1$ | $v_1$ |
| $K_2$ | $v_2$ |
| $K_3$ | $v_3$ |
| $K_4$ | $v_4$ |

Figure 2: One component per face. Component $K_i$ belongs to the face $F_i$ opposite the vertex $v_i$. Deleting $K_k$ leaves the three components belonging to the three faces that meet at $v_k$, so the four instances of condition (i) correspond to the four vertices, and the symmetry group permutes them transitively.

## 5 The symmetry is compatible with the required linking

Before constructing anything it is worth checking that conditions (i) and (ii) are not in conflict, since a symmetry constrains the invariants of a link and could in principle force them to vanish. They are not in conflict, and the reason the check is clean is the following.

**Lemma 5.1.** *The components of a* T*-symmetric four-component link may be oriented so that every* $g \in \mathrm{T}$ *preserves the orientation of every component it maps to itself, and carries chosen orientations to chosen orientations throughout.*

*Proof.* An equivariant choice of orientations is a section of the associated $\{\pm 1\}$-bundle over the component set $\mathrm{T}/\langle r_1 \rangle$, and such bundles are classified by $\mathrm{Hom}(\langle r_1 \rangle, \{\pm 1\})$. Since $\langle r_1 \rangle$ has order 3, which is odd, that group is trivial, so the bundle is trivial and a section exists. □

With orientations so chosen, every $g \in \mathrm{T}$ is an orientation-preserving diffeomorphism of $\mathbb{R}^3$ carrying oriented components to oriented components, so for $\sigma = \rho(g)$,

$$\bar{\mu}(\sigma i, \sigma j, \sigma k) = \bar{\mu}(i, j, k). \tag{3}$$

**Proposition 5.2.** *Let* $L$ *satisfy (ii), with all pairwise linking numbers zero. Then the vector*

$$\big(\bar{\mu}(234),\ \bar{\mu}(134),\ \bar{\mu}(124),\ \bar{\mu}(123)\big)$$

*lies in a one-dimensional subspace of* $\mathbb{R}^4$*, spanned by* $(-1, 1, -1, 1)$*. In particular condition (i), which forces* $|\bar{\mu}(ijk)| = 1$ *on all four triples, is compatible with (ii), and it determines the four signs up to a single global sign.*

*Proof.* Since $\bar{\mu}$ is alternating it is determined by its values on the four three-element subsets written in increasing order, which are the four coordinates above. Equation (3) for $\sigma$ ranging over $\rho(\mathrm{T}) = A_4$ is a system of linear equations in those four unknowns. Take $\sigma = (123)$. Applying (3) to the triples $(1, 2, 4)$, $(2, 3, 4)$ and $(3, 1, 4)$ in turn, and using that $\bar{\mu}$ is alternating to re-sort indices, gives $\bar{\mu}(234) = \bar{\mu}(124)$ and $\bar{\mu}(134) = -\bar{\mu}(124)$. Taking $\sigma = (234)$ and arguing the same way gives $\bar{\mu}(123) = -\bar{\mu}(124)$. Since $(123)$ and $(234)$ generate $A_4$, these are all the relations, and the solution space is spanned by $(1, -1, 1, -1)$ in the ordering above, equivalently by $(-1, 1, -1, 1)$. □

## 6 The link

**Theorem 6.1.** *Let*

$$p = (3, -1, -1), \qquad q = (-1, 0, -2),$$

*let $K_1$ be the closed hexagon through*

$$p, \quad q, \quad r_1(p), \quad r_1(q), \quad r_1^2(p), \quad r_1^2(q)$$

*in that cyclic order, and let $K_i = u_i(K_1)$ for $i = 2, 3, 4$ as in (2). Then $L = K_1 \sqcup K_2 \sqcup K_3 \sqcup K_4$ is a four-component link with all twenty-four vertices in $\mathbb{Z}^3$, it is* T*-symmetric with $\rho$ injective, all six of its pairwise linking numbers vanish, and each of its four three-component sublinks is the Borromean rings.*

Written out, with $r_1(x, y, z) = (z, x, y)$, the four hexagons are the rows of Table 1, and Figure 3 shows the link.

| component | vertices |
|---|---|
| $K_1$ | $(3,-1,-1)$ $(-1,0,-2)$ $(-1,3,-1)$ $(-2,-1,0)$ $(-1,-1,3)$ $(0,-2,-1)$ |
| $K_2$ | $(3,1,1)$ $(-1,0,2)$ $(-1,-3,1)$ $(-2,1,0)$ $(-1,1,-3)$ $(0,2,1)$ |
| $K_3$ | $(-3,-1,1)$ $(1,0,2)$ $(1,3,1)$ $(2,-1,0)$ $(1,-1,-3)$ $(0,-2,1)$ |
| $K_4$ | $(-3,1,-1)$ $(1,0,-2)$ $(1,-3,-1)$ $(2,1,0)$ $(1,1,3)$ $(0,2,-1)$ |

Table 1: The twenty-four vertices of $L$. Each row is a closed hexagon, listed in cyclic order; $K_2, K_3, K_4$ are the images of $K_1$ under the half-turns $u_2, u_3, u_4$ about the three coordinate axes.

*Proof of the first three assertions.* Every element of T used above is a signed permutation of coordinates, so it maps $\mathbb{Z}^3$ to $\mathbb{Z}^3$ and all twenty-four vertices are integral. The curve $K_1$ is $r_1$-invariant by construction, since $r_1$ shifts the listed cyclic order by two places; the four curves are pairwise disjoint, the closest approach between two of them being $0.3123\ldots$ against a diameter of $6.63\ldots$. So Lemma 4.1 applies and gives T-symmetry with $\rho$ injective.

For the linking numbers, choose any direction of projection in which the image is a diagram, compute the finitely many crossings between distinct components, and sum their signs; all the arithmetic is exact because the vertices are integral. Each of the six sums is zero. □

The remaining assertion is the identification of the sublinks, and here our argument is computational.

**Proposition 6.2.** *Each three-component sublink of the link $L$ of Theorem 6.1 has all three pairwise linking numbers zero and has the same Jones polynomial [2, 3] as $\mathcal{B}$. Consequently each is $\mathcal{B}$.*

*Verification.* By Lemma 4.2 the four sublinks form a single orbit under T, so they are equivalent to one another and it suffices to treat one; we verified all four independently as a check. For each, a projection direction was chosen by search, the resulting diagram was simplified using Reidemeister moves of the first and second kinds, and the Kauffman bracket of the reduced diagram was expanded and normalised by the writhe. Each sublink projects to an 18-crossing diagram that reduces to 8 crossings, with the Jones polynomial

$$-A^{-12} + 3A^{-8} - 2A^{-4} + 4 - 2A^4 + 3A^8 - A^{12},$$

in Kauffman's variable $A$, which is the Jones polynomial of $\mathcal{B}$; and the three pairwise linking numbers are zero. Among three-component links of small crossing number, this combination of invariants occurs only for $\mathcal{B}$ [5]. The computations are deterministic and are reproduced by the scripts accompanying this paper. □

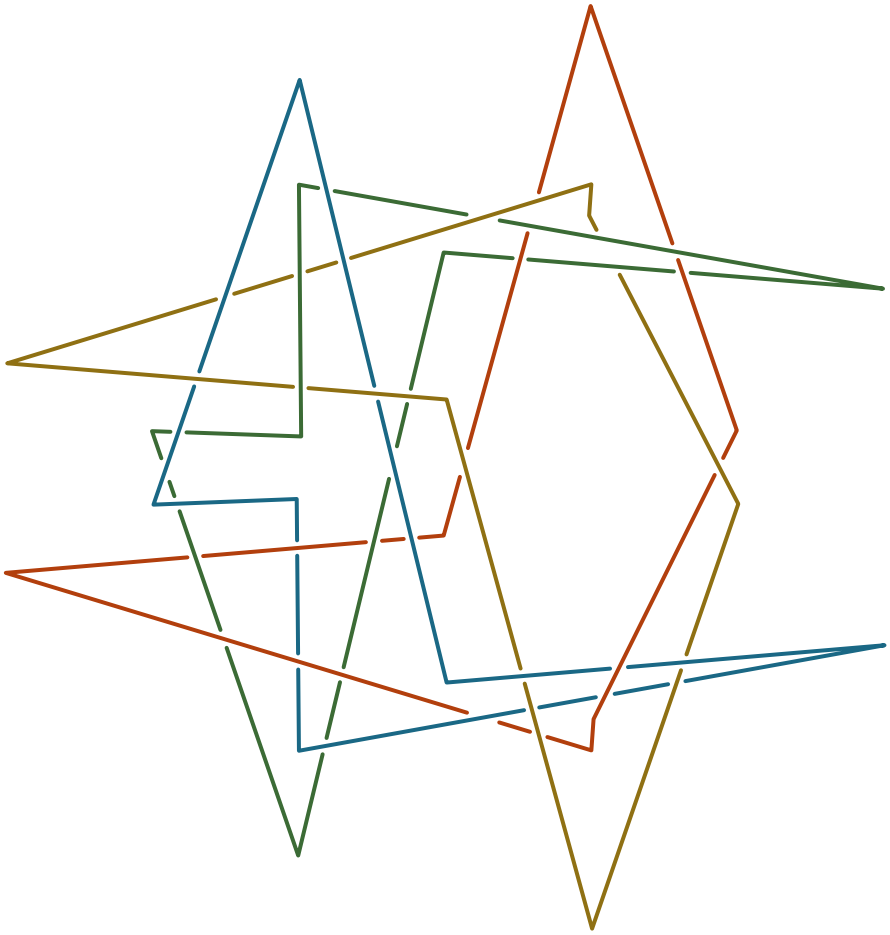


Figure 3: The link $L$ of Theorem 6.1. The projection direction is chosen to keep strands apart: among directions giving close to the fewest crossings, the one maximising the least separation between crossings, between a crossing and a corner, and between two strands that do not cross. Two extra crossings are accepted in exchange, which nearly triples that separation. Components are coloured $K_1$, $K_2$, $K_3$, $K_4$.

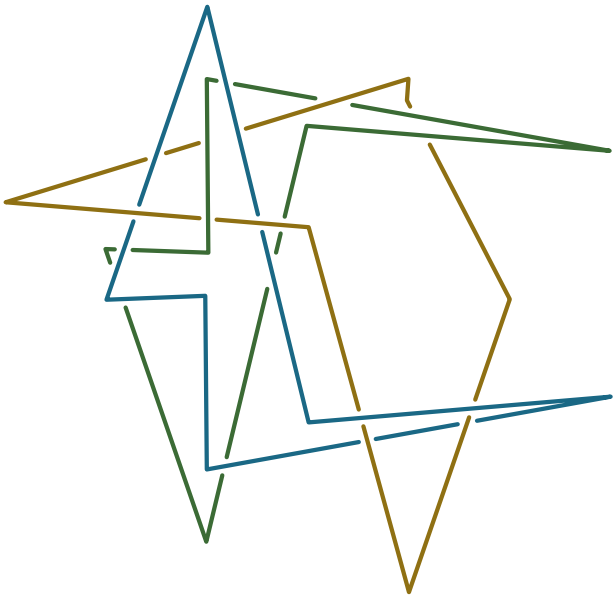


Figure 4: The sublink obtained by deleting $K_1$. By Lemma 4.2 the three surviving components are those belonging to the faces meeting at $v_1$, and by Proposition 6.2 they form the Borromean rings. The other three sublinks are carried onto this one by the symmetry.

## 7 Remarks

**What is and is not proved.** The symmetry in Theorem 6.1 is exact and is established by Lemma 4.1 rather than measured; the vanishing of the pairwise linking numbers is a finite exact computation. The identification of the sublinks as $\mathcal{B}$, however, rests on Proposition 6.2, which appeals to the tabulation of links of small crossing number. A synthetic argument — an explicit isotopy carrying one sublink onto the standard diagram of $\mathcal{B}$, which by Lemma 4.2 would settle all four at once — would be preferable, and we do not have one. The natural route is to redesign the component so that the Borromean weave is visible in a neighbourhood of each vertex and the connecting arcs demonstrably meet nothing.

**Acknowledgement.** The observation that the four components should be indexed by the faces of a tetrahedron, so that triples of components correspond to vertices, is due to Kyle Miller [6].

## References

[1] Michael H. Freedman and Richard Skora. Strange actions of groups on spheres. *Journal of Differential Geometry*, 25(1):75–98, 1987.

[2] Vaughan F. R. Jones. A polynomial invariant for knots via von Neumann algebras. *Bulletin of the American Mathematical Society (N.S.)*, 12(1):103–111, 1985.

[3] Louis H. Kauffman. State models and the Jones polynomial. *Topology*, 26(3):395–407, 1987.

[4] Bernt Lindström and Hans-Olov Zetterström. Borromean circles are impossible. *The American Mathematical Monthly*, 98(4):340–341, 1991.

[5] Charles Livingston and Allison H. Moore. LinkInfo: Table of link invariants. `https://linkinfo.sitehost.iu.edu`. Accessed 2026-10-12.

[6] Kyle Miller. Answer to "Is there a 4-component link such that upon removing any one of them you get the Borromean link?". Mathematics Stack Exchange, 2018. https://math.stackexchange.com/a/2992328, accessed 2026-10-12.

[7] John Milnor. Link groups. *Annals of Mathematics (2)*, 59(2):177–195, 1954.

[8] John Milnor. Isotopy of links. In R. H. Fox, D. C. Spencer, and A. W. Tucker, editors, *Algebraic Geometry and Topology: A Symposium in Honor of S. Lefschetz*, pages 280–306. Princeton University Press, Princeton, NJ, 1957.